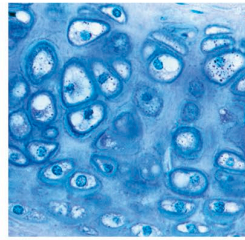


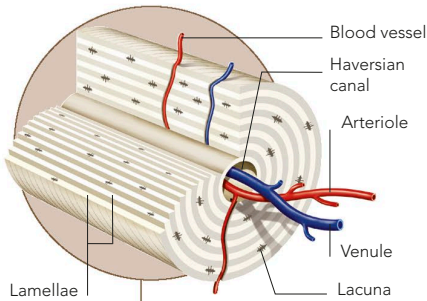
CARTILAGE

Cartilage is a tough, adaptable form of connective tissue. It consists of a gel-like matrix containing many chemicals, such as proteins and carbohydrates. In this are embedded various types of fibres, and cells called chondrocytes, which make and maintain the whole tissue. There are several kinds of cartilage, including hyaline cartilage, fibrocartilage, and elastic cartilage, a springy material found at sites such as the outer ear flap and larynx.



HYALINE CARTILAGE

Dense collagen fibres make this cartilage extra tough and resistant. It covers bone ends in joints, attaches ribs to the sternum, and is also found in the trachea and nose.



OSTEON

This rod-shaped unit is the building block of compact bone. Its central (Haversian) canal, containing blood vessels and nerves, is surrounded by concentric layers of tissue (lamellae). Gaps (lacunae) in the tissue contain osteocytes, which maintain bones.

Bone marrow

Tissue filling a bone's central cavity; at first, long bones have red marrow – later this turns into yellow marrow

Osteon

Vein

Artery

Spongy bone

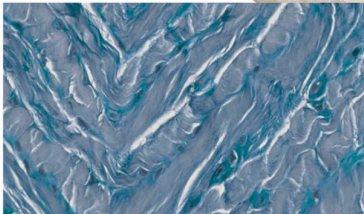
Latticework structure consisting of bony spikes (trabeculae), arranged along lines of greatest stress

Epiphysis

Expanded head of bone containing mainly spongy bone tissue

Bone shaft

Mostly compact bone and marrow



FIBROCARILAGE

This is mostly dense bundles of collagen fibres, with little gel-like matrix. It is found in the jaw, knee joints, and intervertebral discs.